Fall 2026 - PHYS 5350

UCONN
COLLEGE OF LIBERAL ARTS AND SCIENCES

COURSE DETAILS

Course Title: Computational Physics
Lecture: T/R 5:00 PM - 6:15 PM   **Location:** GS 119
Professor: Dr. Aleksandra Kuznetsova (*she/her*)
Email: aleksandra.kuznetsova@uconn.edu
Office Hours: T/R 4:00 PM - 5:00 PM   or by appointment (link on HuskyCT)
Office: Gant South 113H

ABOUT ME:

Prof/Dr. Kuznetsova (koos-neh-TSOH-vah), please call me Aleksandra. I am a computational astrophysicist. Outside of class, I spend most of my time working with my research group, the Origins Group, here at UConn, where we use numerical methods to study the birth of stars and planets in our Galaxy.

CONTACT INFO

During the day, the easiest way to get in touch is to stop by my office (Gant South 113H). I am guaranteed to be available during my listed office hours, but I am generally available if my door is open. You may additionally schedule a time to come by my office if you need to get in touch outside of class or office hours, by the link on HuskyCT which will book you directly on my calendar. To contact me asynchronously, you can email me. I typically check my email on weekdays between 9 AM and 5 PM and will try to respond to questions within 24-48 hours.

COURSE MATERIALS

- Textbook (**required**): *Computational Physics*, Mark Newman, 2012, 1st Ed. Most of the assigned readings for this course will come out of this textbook.
- Laptop (**required**): Access to a laptop computer will be required to complete the coding exercises during class time, the homework assignments, and the final project. Please, let me know if you do not have access to a suitable laptop computer so that we can discuss options as soon as possible.
- Course Website (**required**): https://akuznetsova.github.io/2026_compphys. Lecture and interactive in-class content, guidelines for assignments, and supplemental materials for the problem-sets are hosted on the course website. The website will be updated weekly with new materials.
- Python (**required**): We will use the Python programming language throughout this course. You should have a working Python distribution (3.+) on your laptop. For in-class exercises, you will also need Jupyter lab.
- Package/Environment manager (*recommended*): Will help you use external libraries and packages. A popular choice is Anaconda which will manage the installation of a python distribution alongside relevant scientific libraries. See the conda documentation for a quickstart guide.
- LaTeX (**required**): Your final project should be submitted as a PDF document created in LaTeX. You can open an account and run LaTeX online at `https://www.overleaf.com.`. Alternatively, you can also install LaTeX locally on your computer. If you are new to LaTeX, you can find many tutorials on overleaf.

Course Objectives

PHYS 5350 is a graduate-level introduction to computational physics. The overall course learning goals are for students to be able to:

1. Understand the basic principles and evaluate the limitations of core numerical algorithms and techniques used in the solution of physics problems.
2. Apply computer programming best practices to write code in Python for scientific computing.
3. Demonstrate proficient communication (verbal and written) of the methodology and results of numerical models with scientific visualization techniques.

Course Schedule and Topic Outline[1]

[1] tentative dates, please regularly check the website regarding modifications of deadlines

Note: Assigned readings should be completed *prior* to the start of the week listed in the table.

Week	**Dates [T,R]**	**Topics**	**Reading [Newman]**	**Notes**
1	09.01, 09.03	course overview, python overview	Ch. 2	
2	09.08, 09.10	visualization, other tools	Ch. 3	W: PS1 due
3	09.15, 09.17	accuracy	Ch. 4	W: PS2 due
4	09.22, 09.24	integration, differentiation	Ch. 5	F: *R&R* (PS1/PS2)
5	09.29, 10.01	linear and non-linear equations	Ch. 6	W: PS3 due
6	10.06, 10.08	ordinary differential equations (ODEs)	Ch. 8.1-8.4	W: PS4 due
7	10.13, 10.15	ODEs: additional methods	Ch. 8.5-8.6	F: *R&R* (PS3/PS4)
8	10.20, 10.22	partial differential equations (PDEs)	Ch. 9	W: PS5 due
9	10.27, 10.29	PDEs	Ch. 9	W: project pitch due; F: *R&R* (PS5)
10	11.03, 11.05	fourier transforms	Ch. 7	W: PS6 due
11	11.10, 11.12	Monte Carlo methods	Ch. 10	W: PS7 due
12	11.17, 11.19	Monte Carlo methods	Ch. 10	W: project abstract due; F: *R&R* (PS6/PS7)
13	11.24, 11.26	*Thanksgiving Break*		*no class*
14	12.01, 12.03	selected topics	handouts	W: PS8 due
15	12.08, 12.10	final presentations		F: *R&R* (PS8)
Finals		*no final exam*		M: final projects due

R&R: deadline to revise and resubmit any problems from the indicated problem sets.

Course Format

Students can expect to engage with the course material in the following ways:

1. Assigned reading from the textbook (and supplemental notes on the course website), prior to class
2. Lectures and interactive examples (during each class)
3. In-class (individual and group) coding exercises and discussion
4. Take-home problem sets – eight (8) over the course of the semester.

The course is graded on a non-traditional scheme where the final grade is based on achieving the minimum required scores **across all three (3) columns** of performance based on the following scale:

	In-class work (weekly)	Problem-Sets (by problem)	Final Project (paper/presentation)
A	6 $\checkmark+$, 4 $\checkmark$	50 % $\checkmark+$, 40 % $\checkmark$	$\checkmark+$, $\checkmark$
A-	5 $\checkmark+$, 5 $\checkmark$	40 % $\checkmark+$, 50 % $\checkmark$	$\checkmark+$, $\checkmark$
B+	4 $\checkmark+$, 6 $\checkmark$	20 % $\checkmark+$, 70 % $\checkmark$	$\checkmark$, $\checkmark$
B	3 $\checkmark+$, 7 $\checkmark$	90 % $\checkmark$	$\checkmark$, $\checkmark$
B-	3 $\checkmark+$, 6 $\checkmark$	80 % $\checkmark$	$\checkmark$, $\checkmark$
C+	2 $\checkmark+$, 6 $\checkmark$	70 % $\checkmark$	$\checkmark-$, $\checkmark$
C	1 $\checkmark+$, 6 $\checkmark$	60 % $\checkmark$	$\checkmark-$, $\checkmark$
D	6 $\checkmark$	50 % $\checkmark$	$\checkmark-$, $\checkmark-$

Weekly In-Class Exercises

Programming is the kind of skill that is best learned by doing. Time will be designated in class to working individually and in groups to solve problems, do some hands-on coding, and having class-wide discussions. Students will submit their work on the week's in-class problems each week by noon on Fridays on HuskyCT. Late in-class work will generally only be accepted up until one (1) week after the deadline and can earn no more than a $\checkmark$.

Scoring Criteria:

$\checkmark+$: Work is fully complete and correct.

$\checkmark$: Work shows understanding, but still has some errors.

$\checkmark-$: Reasonable start, but work is sufficiently inaccurate or incomplete.

Problem-Sets

Take home problems are intended to challenge you to go beyond the material presented in class and exercise your programming problem-solving skills. Collaboration with other class members is allowed and encouraged, though everyone is expected to turn in their own copy of the assignment. Problem sets will be assigned at least a week ahead of their due date and should be submitted by 11:59 PM on the day of the deadline according to the submission instructions on HuskyCT and the information provided on the course website. Each individual problem receives a score of $\checkmark+$, $\checkmark$ or $\checkmark-$.
Revise & Resubmit: After the submission deadline, students will have access to a report with the correct output of the code. Up until the *R&R* deadline for the problem set, individual problems may be corrected and re-submitted for a potential one level improvement up the scale e.g. $\checkmark- \to \checkmark$ or $\checkmark \to \checkmark+$.

Final Project

To apply and demonstrate the numerical skills we are working on, you will be expected to pitch, develop, and execute an idea for the application of a relevant numerical method to a physics problem of your choice. Projects will receive a score for the written report and another score for a short final presentation delivered to the class during the last week of the semester. More details to follow.

Late work and Deadlines

Late problem sets are effectively treated as resubmissions, with a default score of $\checkmark-$ on the first round, such that the potential maximum score for a late problem is a $\checkmark$. If you think you may need an extension on an assignment, you may request one using the form on HuskyCT. Extensions should be requested *in advance of the deadline.*

Missing Class

You may need to miss class on occasion and that is okay. The grading scheme is designed such that you can miss a few classes without severe penalty to your grade. If you find that you need to miss class, you can take the following steps to ensure you're caught up:

To take care of yourself and others around you, please stay home if you are experiencing symptoms of illness.

1. Make sure you have done the assigned reading listed on the syllabus for that class period.
2. Check the course website to review the lecture material and in-class exercises you missed.
3. Complete the in-class exercises on your own time; you may wish to ask a colleague to help get you up to speed.
4. Attend office hours to discuss any content or questions you have regarding the missed material or make-up work.

Help me help you!
If you are

- missing class for an extended period of time (> 2 weeks),
- are regularly missing 1 out of every 2-3 classes,
- or otherwise find yourself in a situation in which you are not able to participate/engage/stay on top of the course material

please get in touch *as soon as possible.* Accommodations can be made, but I can't help if I don't know about the problem!

For the routine or occasional absence, you don't need to email the instructor or inform them of your absence. To find out what you missed, please consult the above procedure.

Course Policies

Please aim to restrict the use of laptops in class to note-taking and working on the in-class exercises. Minimize distractions for yourself and your fellow students; please silence notifications on your electronic devices (cell phones, tablets, laptops etc.) and put away devices when not in use for note-taking or in-class work.

Collaboration

Physics, like many sciences nowadays, is done in collaborative environments. Working in a group with others is a big part of how science gets done and learning how to be an effective collaborator is one of the key skills we will practice in classes like this one. In class, students are expected to work effectively with others [2], participate in discussions, and practice taking on different roles in their groups. The most effective scientific communities tend to be those in which all members are able to communicate efficiently with one another, but building environments in which members feel welcome and safe to contribute doesn't happen overnight or on accident. Please, aim to be respectful and inclusive of your colleague's contributions, ideas, and experiences.

[2] *All members of the University community are responsible for the maintenance of an environment in which people are free to learn and work without fear of discrimination, discriminatory harassment or interpersonal violence.*
– UConn Discrimination Policy

Academic Integrity

Academic honesty and integrity is a fundamental tenet of education and research: academic work depends on respect for and acknowledgment of the research and ideas of others. Misrepresenting someone else's work as your own is plagiarism – a serious offense. Plagiarism includes copying solutions from other students, external resources, as well as the use of generative AI. As a student, it is your responsibility to avoid plagiarism and meet University expectations related to academic integrity. [3]

Beyond that, I want you to be able to be proud of your work and intellectual accomplishments. Learning physics, and how to do science in general, is often challenging but it is in the process of learning and reasoning that we get to experience the joy of making sense of something in the world.

[3] All students are expected to act in accordance with the Guidelines for Academic Integrity at the University of Connecticut.

Generative AI and the use of External Resources [4] [5]

The efficiency gains of generative AI within programming remains a contested claim. Regardless, in this course, programming is taught as process. That is, the value is in the journey rather than the destination. Using generative AI to generate code sidesteps the process and curtails the foundational skill-building which is the ultimate aim of a course like this one. In addition to being a *violation of academic integrity in this course*, the use of generative AI is antithesis to the scientific process: by definition, this software cannot reason, evaluate, accurately source, or verify its output. *The use of uncited (and thus unverifiable) information is incredibly poor scientific practice.*

Research practice depends on the use of external resources. Please, keep in mind, when you submit any written work that external resources must:

(1) come from reliable and *credible* secondary or primary sources [6] and

(2) be appropriately cited and credited *in writing*. [7]

[4] generative AI includes but is not limited to: large language models (LLMs) such as chatGPT or Claude, built-in assistants like co-pilot, google's Gemini and similarly "AI-powered" search output, etc.

[5] The environmental costs of powering and cooling the data centers used for AI models are already unsustainable and have an outsized impact on the most vulnerable communities. Please, do not use generative AI in this course.

[6] e.g. encyclopedias, textbooks, materials from educational institutions, communication from scientific agencies like NASA, NRAO, NOAA, etc., published articles from scholarly publications, etc.

[7] See resources from the Online Writing Lab and Scientific Style and Format Guide for guidance and examples

Additional Resources

Students who require formal accommodations are encouraged to contact the Center for Students with Disabilities to make arrangements for the semester, as soon as possible. This lets me know what your access needs are so I can try to make sure they are being met!

If you'd like to discuss any additional ways you could be better supported in the course, please do not hesitate to reach out to me.

The University of Connecticut is committed to protecting the rights of individuals with disabilities and assuring that the learning environment is accessible.

Student Support Services

- Counseling and Mental Health Services: cmhs.uconn.edu
  Phone: (860) 486-4705 (after hours, use (860) 486-3427)
- Career Services: career.uconn.edu
  Phone: (860) 486-3013
- Alcohol and Other Drug Services: aod.uconn.edu
  Phone: (860) 486-9431
- Office of Student Services & Advocacy: ossa.uconn.edu
  Phone: (860) 486-3426
- Technology Support Center: techsupport.uconn.edu
  Phone: 860-486-4357
- Center for Students with Disabilities: csd.uconn.edu
  Phone: (860) 486-2020 (voice), (860)486-2077 (TDD)